

How EAE lesions *form, resolve or persist* — and what the segmentation stains add

Mouse EAE spinal cord, RRMAP2 runs 1–6 (all five runs): 67 animals, 1.38 M annotated cells, daily clinical scores, four segmentation-kit stains imaged for every section. Chronic (MOG35–55) and relapsing–remitting (PLP139–151) courses. Analyses in [notebooks/10–16](#); illustrated walk-through in [notebooks/12_lesion_story.ipynb](#).

Lesions run from an *active infiltrate* to a *fibrotic, demyelinated* state and largely resolve. In animals that do well, lesions fill with *vimentin-bright reactive astrocytes*, which only the images can see. Severe chronic animals keep active lesions without it, and relapse-prone animals accumulate meningeal B-cell aggregates.

- **Lesions redefined.** Against control tissue, never-immunised animals come out ~1 % lesion, down from 4–20 % with the curated labels. The lesion share falls from 0.93 at the first peak to 0.14 in long remission.
- **Six lesion states:** active monocyte-derived (S1, S4) at every peak; fibrotic / myelin-loss (S2) and glial-reactive (S0, S3) later; lymphocytic infiltrate (S5) throughout.
- **Vimentin** (*Vim* is not on the panel): lesion astrocytes carry more vimentin in milder / resolving disease in every contrast. It is the strongest correlate of chronic severity ($\rho = -0.80$). It survives batch, brightness, focus and α SMA controls. It fills the lesion interior (no ring at the edge): low at peak and strongest in mild and never-relapsing animals, so it marks **a lesion-wide resolution response**. Within-lesion tests show it does not act as a measurable barrier to infiltrating immune cells.
- **Chronic severity** is set at the first attack (all SEVERE peaked at 3.5) and means lesions that stayed active, not more neuropil loss.
- **Relapse** can't be predicted clinically. B cells accumulate after the first attack and move into meningeal aggregates (fewer in never-relapsing animals). Relapses are mostly new lesions, with some flaring at old borders.
- **Earlier image findings replicate** in runs 1–3: white-matter neuropil loss, astrocyte RNA-before-vimentin, T cells facing vessels, image models on unseen runs. The 18S texture marker replicates only partly.

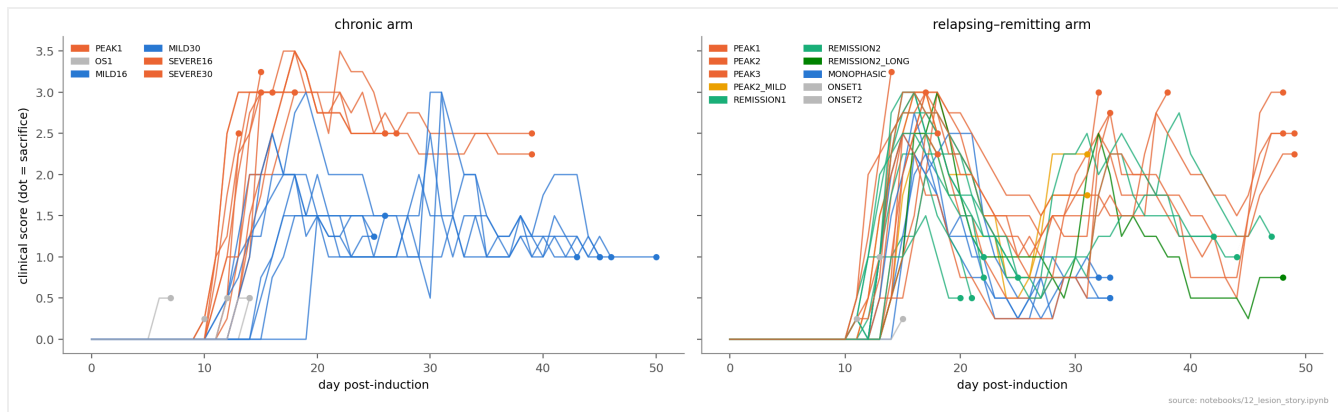


Figure 1 • Disease courses. Daily scores of every animal, coloured by sacrifice stage (dot = sacrifice). Severe chronic animals all peaked at 3.5 and never recovered; first attacks look alike in animals that will and won't relapse.

1 Lesions redefined, six lesion states

Each cell's 30-cell neighbourhood (cell-type mix, seven disease programs, cellularity) is compared with control animals of the same region (white matter, grey matter, meninges). It counts as lesion when it lies outside the control range, calibrated so held-out controls are ~1 % lesion. 72 % of cells labelled "Lesion_mix" look like normal tissue.

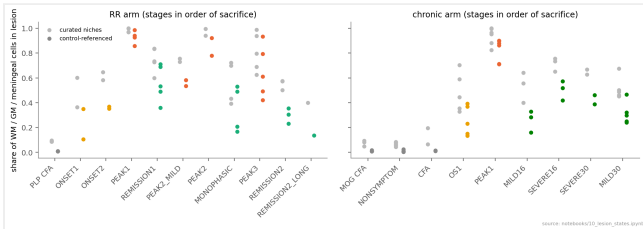


Figure 2a · Lesion share per animal, curated (grey) vs control-referenced.

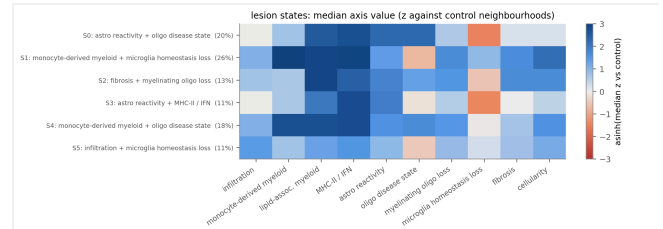


Figure 2b · The six lesion states: median axis value vs control.

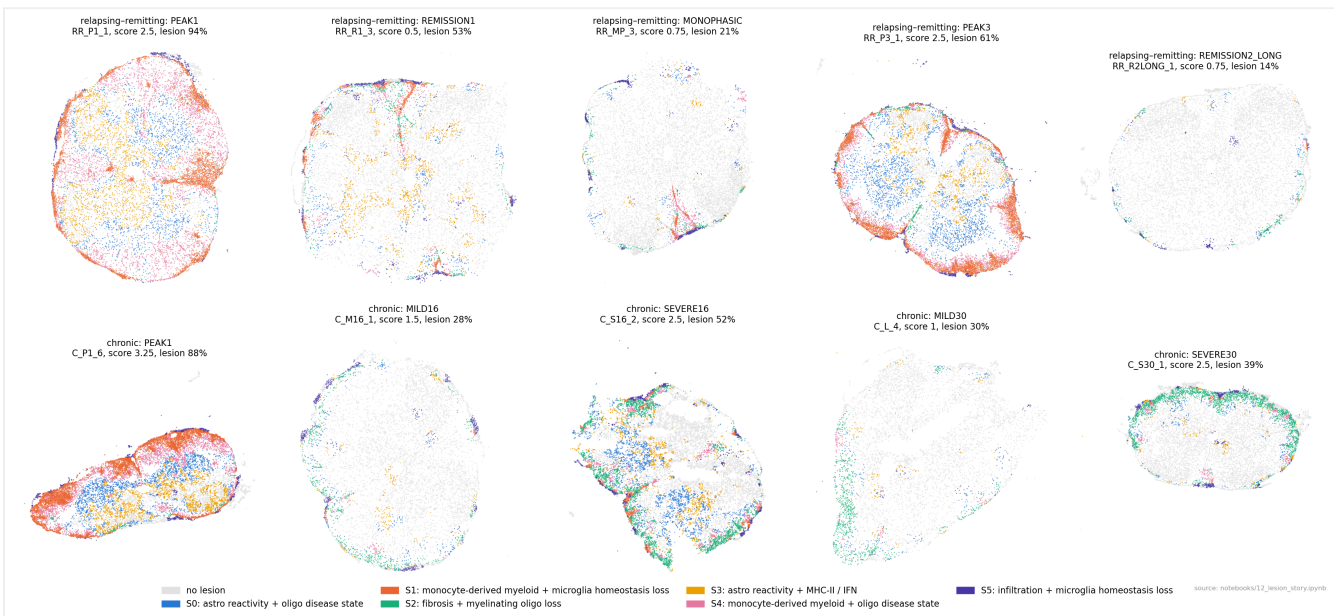


Figure 3 · Lesion states in the tissue, one representative animal per stage (median lesion share, largest piece). Active monocyte-derived lesion (orange/pink) fills the white matter at the first peak. Later, fibrotic (green) and glial-reactive (blue) tissue remains; long remission leaves small residues; chronic severe animals keep active tissue at d28–41.

2 Vimentin marks milder and resolving lesions

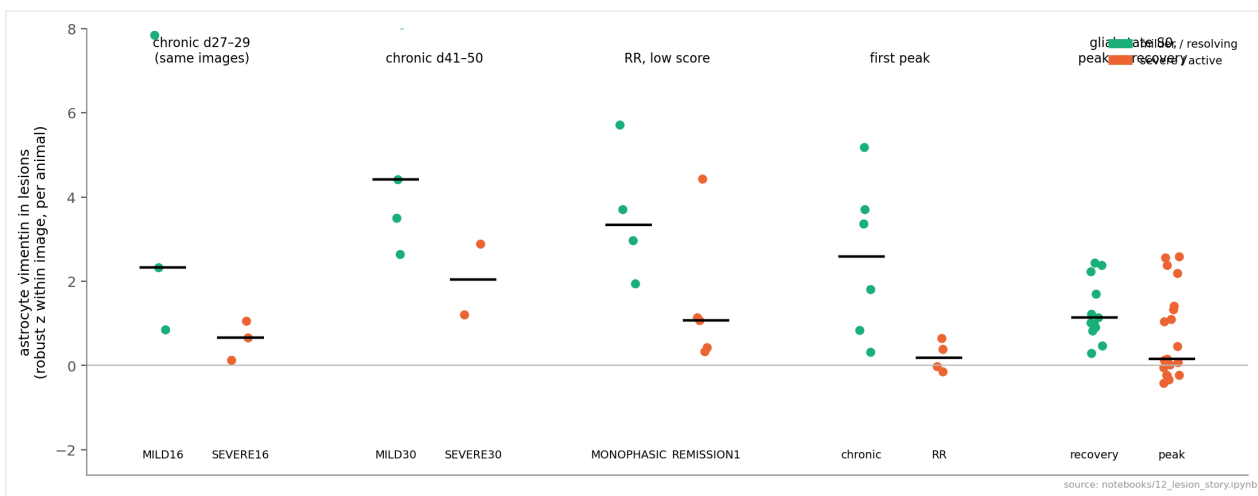


Figure 4 · Lesion astrocyte vimentin per animal in five independent contrasts; green = milder / resolving.

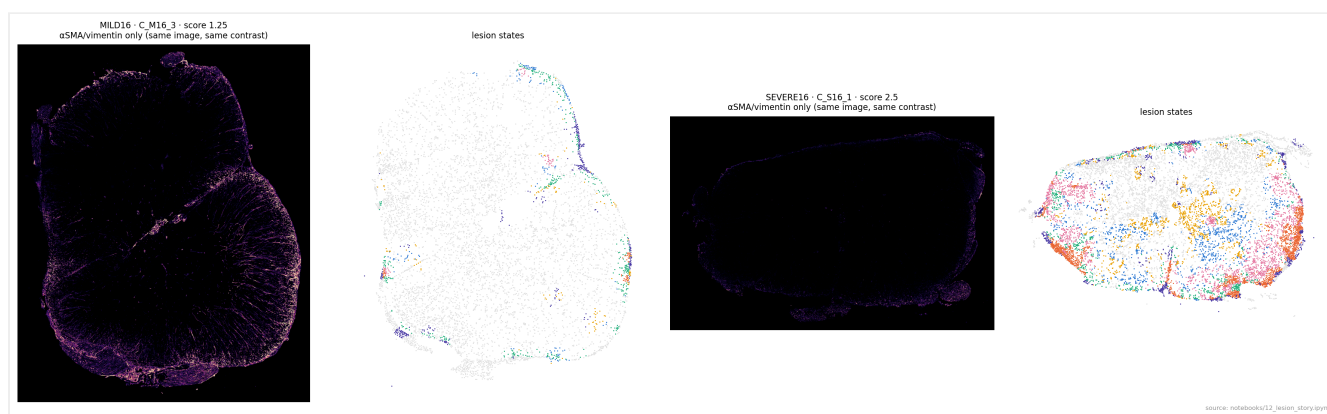


Figure 5 · Same image, same contrast: αSMA/vimentin channel of a MILD16 and a SEVERE16 piece, with lesion states. (The severe piece is dark overall; the quantitative evidence is the within-piece contrast.)

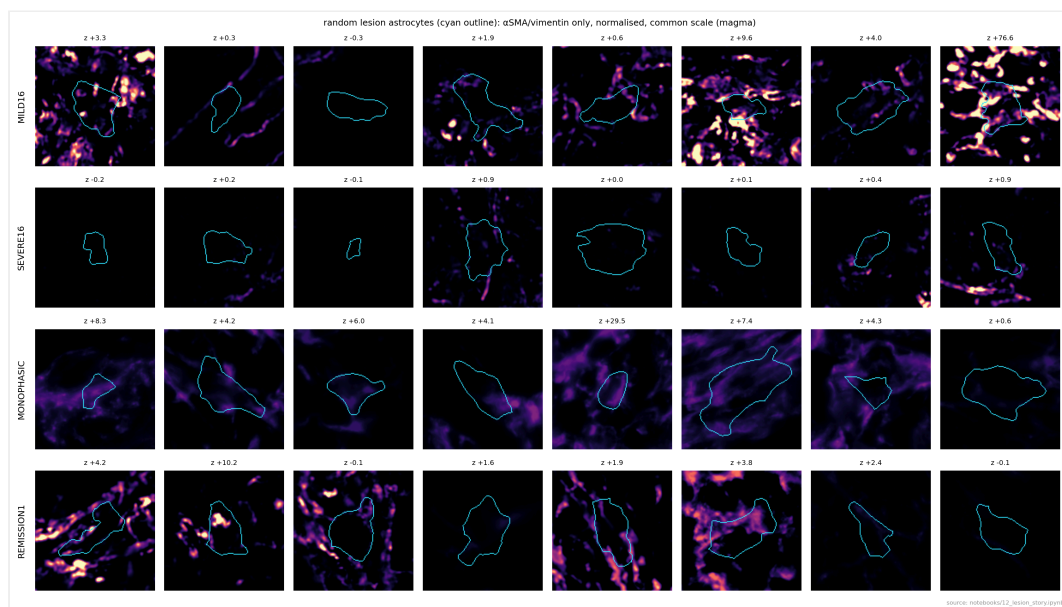


Figure 6 · Random lesion astrocytes (cyan outline), vimentin channel only, normalised per image, common scale.

Technical or αSMA? Every alternative tested (notebook 15)

The effect keeps its direction in all four contrasts within animals, far from vessels, with **no VSMC, endothelial or fibroblast cell within 20 μm**, in deep white matter, in reactive astrocytes only, and within lesion states. Controls stay near zero: DAPI, neurons and oligodendrocytes, focus, and smooth-muscle / pericyte / fibroblast RNA in the astrocytes. A VSMC calibrator flags some channel-brightness contribution in MILD16 vs SEVERE16 and chronic vs RR; the within-animal contrast removes it.

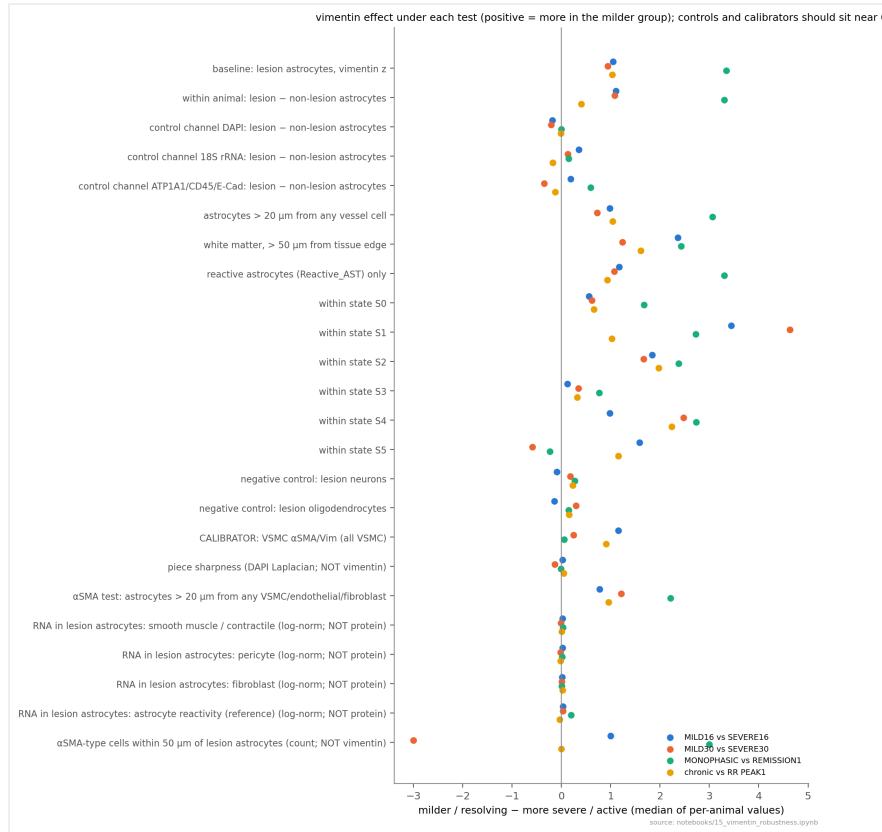


Figure 7 · Robustness. Each row a test, each colour a contrast; positive = more vimentin in the milder group.

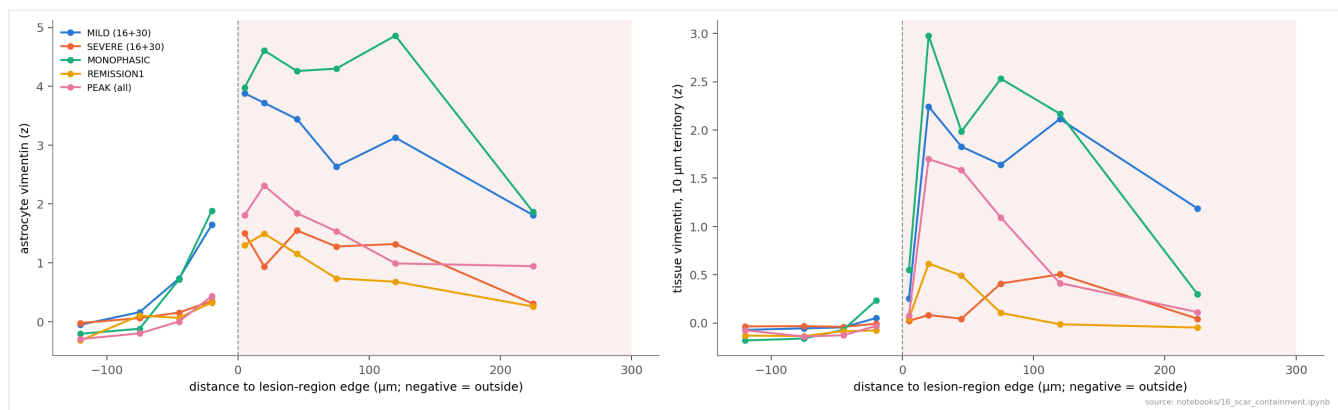


Figure 8 · Where in the lesion. Vimentin against distance to the edge of lesion regions (real outlines): it rises on entering the lesion and stays high through the interior, highest in MILD and MONOPHASIC animals. Inside – outside astrocyte vimentin: PEAK1 0.60; MILD16 2.27 vs SEVERE16 1.01; MILD30 3.30 vs SEVERE30 2.07; MONOPHASIC 4.28 vs REMISSION1 1.26. (An earlier "edge ring" came from a cell-based edge distance and is withdrawn.)

3 Does a vimentin scar contain lesions?

A collaborator's hypothesis: a successful astrocyte scar restricts immune infiltration. With lesion **regions** (smoothed lesion-cell density, 356 regions in 53 animals, real outlines, depth from the cord outline), lesions with more border vimentin have **neither fewer nor more** immune cells around them than other lesions of the same animal (median p 0.00, $p = 0.65$; deep regions $p -0.39$, n.s.). Where vimentin is bright it fills the lesion rather than lining it. A first analysis with cell-linkage objects suggested the opposite; its objects were diffuse scatters without real outlines, so it is withdrawn. Testing restriction needs time-resolved data or an intervention on astrocyte reactivity.

Sharper within-lesion tests (notebook 17). Each lesion edge cut into 100 μm stretches (3,188 stretches, 392 lesions, 53 animals) and compared within the same lesion, adjusted for the immune load inside, activity and depth: vimentin-rich stretches do **not** have fewer immune cells just outside (astrocyte vimentin, median per-animal $p +0.035$, 95 % CI -0.016 to $+0.073$; a barrier would give $p < 0$). They have more immune cells inside the edge and the same outside. Vimentin-rich old lesions at relapse have as much new active tissue next to them ($p -0.02$). **Answer at this resolution: vimentin marks the resolution phase; it does not act as a measurable barrier.**

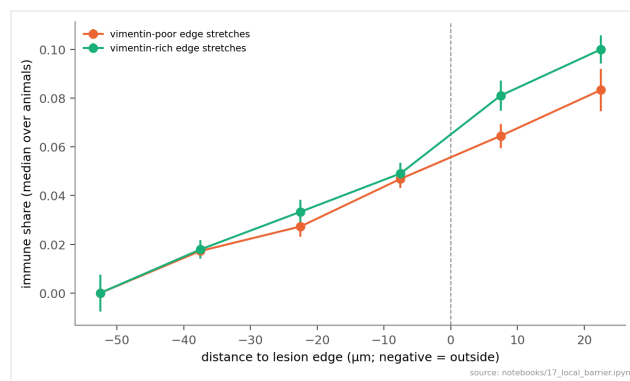


Figure 9b · Immune share across the lesion edge for vimentin-rich vs vimentin-poor stretches of the same lesions: no steeper drop-off across vimentin-rich edges.

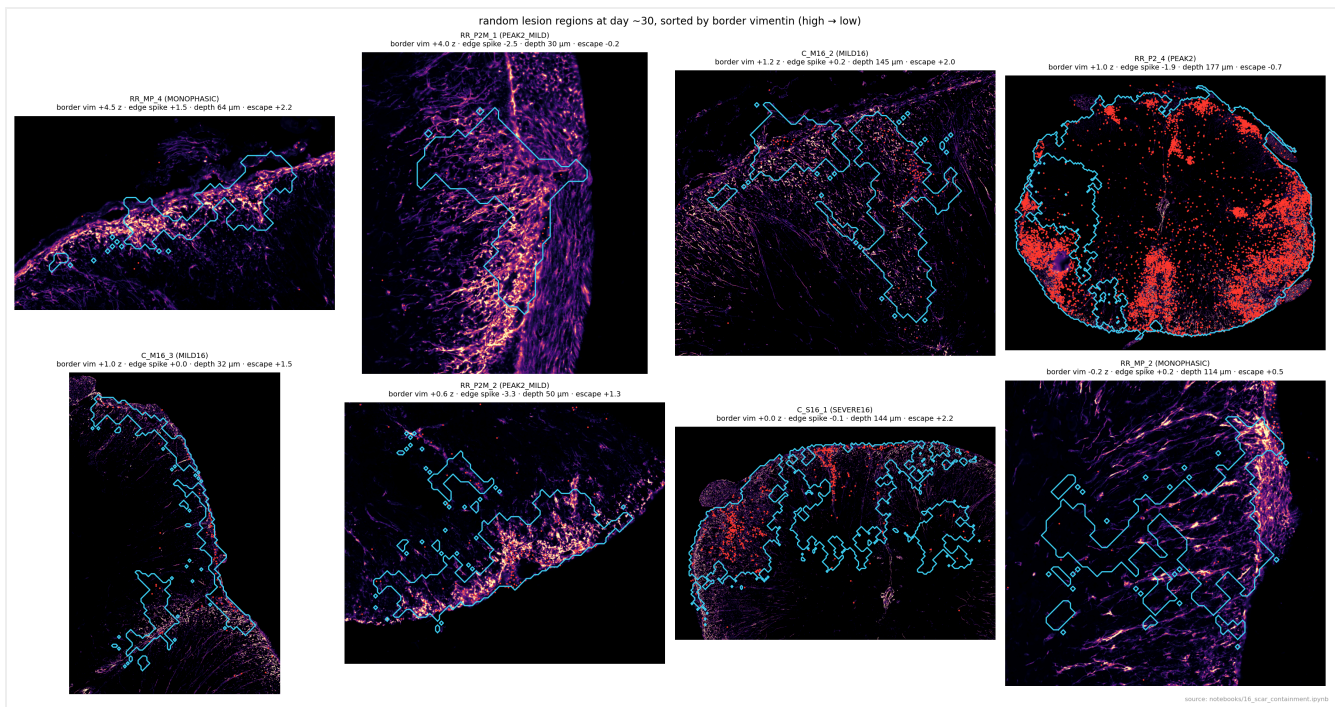


Figure 9 • Random lesion regions at day ~30 (not chosen by eye), sorted by border vimentin: αSMA/vimentin (magma), region outline (cyan), infiltrating immune cells (red).

4 Chronic severity: lesions that stayed active

Severe chronic animals still carry more lesion (0.52 vs 0.28), active monocyte-derived states, T cells and dense infiltrates at d28–41, and fewer myelinating oligodendrocytes (0.08 vs 0.15 per WM cell); neuropil is not lost more. Across 13 chronic-late animals the strongest correlate of the score is **low lesion vimentin** ($p = -0.80$), then lesion share (0.58) and T cells (0.47).

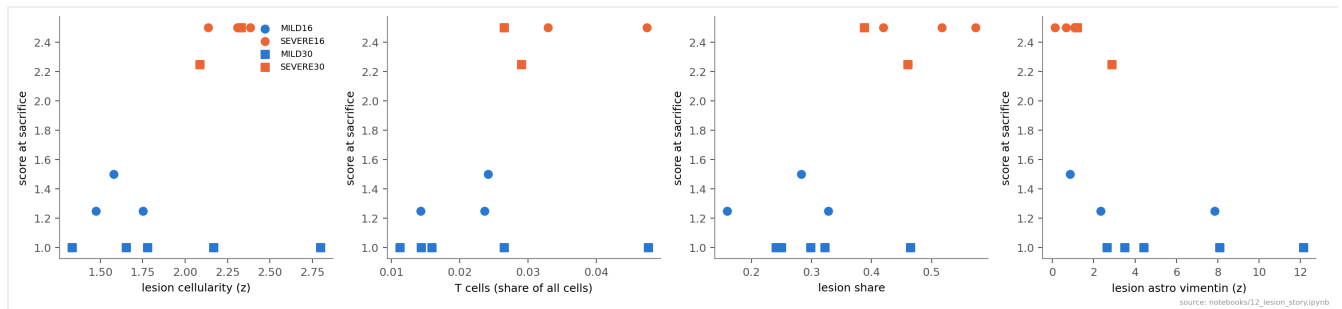


Figure 10 · Score at sacrifice vs lesion measures, chronic-late animals.

5 Relapse: B cells, and new vs reactivated lesions

B cells per 1000 cells: RR PEAK1 14 → REMISSION1 22 → PEAK2_MILD 25; aggregated share 0.26 → 0.55; meningeal share 0.16 → 0.58–0.90. Never-relapsing animals: 9 vs 22 ($p = 0.03$), deeper first recovery, more lesion vimentin. Chronic arm ≤ 8 throughout.

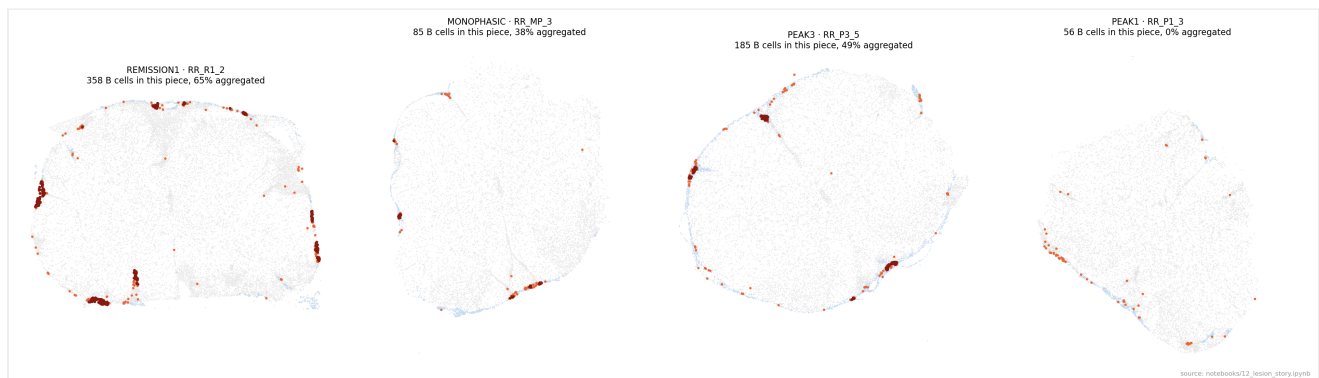


Figure 11 · B cells (red; aggregated dark red) and meninges (blue).

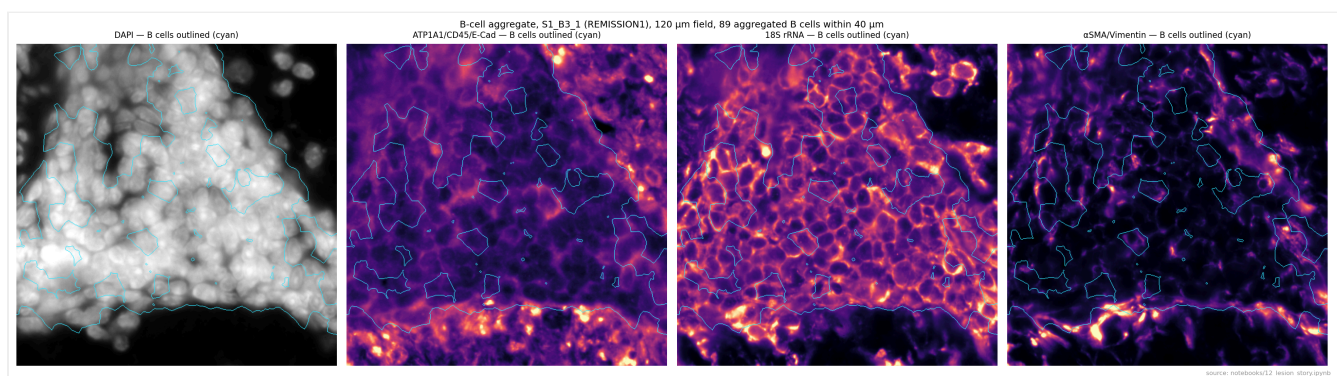


Figure 12 · A B-cell aggregate (REMISSION1), each stain separately, B cells outlined.

Relapse is mostly new lesions, plus flaring at old borders. Active tissue at relapse touches old fibrotic tissue more than at the first attack (0.23 vs 0.05 within 50 μm , $p = 4e-4$). Relapse animals have more

separate lesions (17 vs 8), and only 19 % of active relapse tissue is in lesions that contain old tissue. B-cell aggregates sit near old lesions at relapse (0.72 vs 0.31).

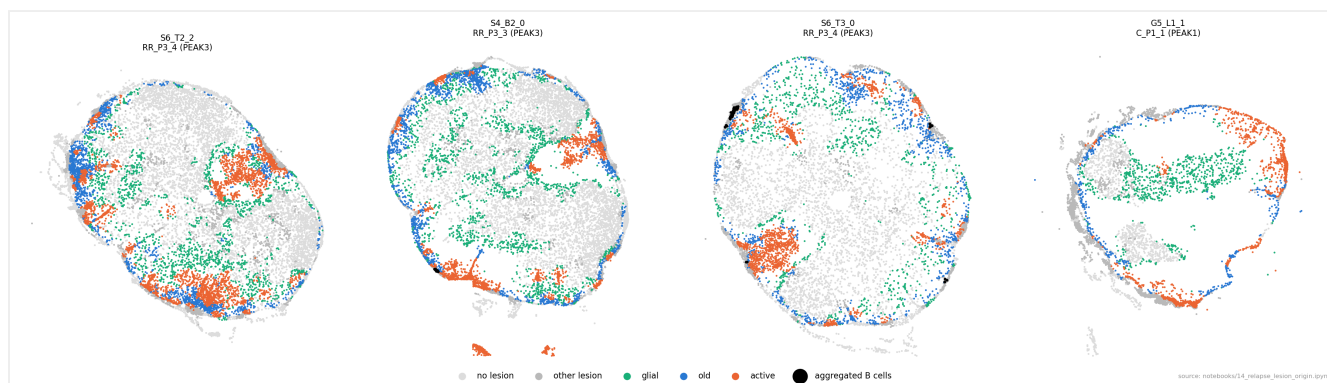


Figure 13 · Active (orange) vs old fibrotic (blue) tissue, B-cell aggregates (black): three relapse pieces and a first-attack piece.

6 Replication, damage memory, caveats

Earlier image finding	Runs 1–3
White-matter neuropil loss in lesions	replicated: $\times 0.72$ (58 pieces, $p < 1e-4$) vs $\times 0.79$
Astrocyte reactive RNA before vimentin protein	replicated: p 0.51–0.58 per run; RNA-only astrocytes more common in active disease
T cells orient 18S-rich cytoplasm towards vessels	replicated: $\cos$ 0.19 (33 animals) vs 0.14
Image-only models on an unseen run	yes, every run: lesion AUROC 0.85–0.91, cell type 0.42–0.47
18S texture severity marker	partly: astrocytes, myeloid, oligodendrocytes only

Damage memory. Within the same lesion state, neuropil trends back towards normal from peak to recovery while vimentin rises (19 vs 13 animals; best $q = 0.13$). Persisting lesions look repaired in neuropil but scarred in vimentin.

Caveats

- Small groups (2–19 animals): the evidence is consistency across independent contrasts, not single p values.
- The arms differ in antigen, likely strain, and run. MILD30 is run 1 only. MONOPHASIC and REMISSION1 never share an image.
- Lesion calls and states come from the transcriptome; the image readouts are independent of them.
- Cross-sectional design: "builds", "precedes" and "resolves" are inferred from animals sacrificed at different times.

Next

1. Validate vimentin vs α SMA with separate antibodies on a few MILD/SEVERE and MONOPHASIC/REMISSION1 sections.
2. Order events along the active $\rightarrow$ fibrotic trajectory (when neuropil, vimentin and 18S change).
3. Time-resolved or interventional test of whether the vimentin border restricts infiltration.